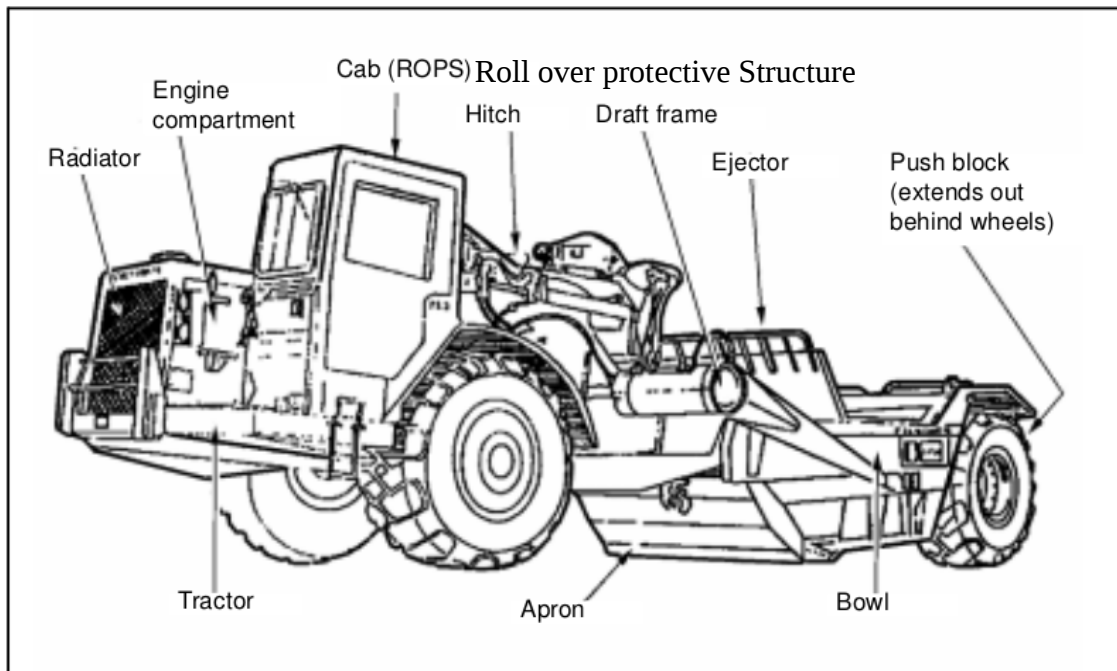


SCRAPERS

The design of scrapers (tractor scrapers) allows for loading, hauling, dumping, and spreading of loose materials. Scrapers are used for medium-haul earthmoving operations and for moving ripped materials and very loose rock. The haul distance (zone of operation), the load volume, and the type and grade of surface traveled on are the primary factors in determining whether to use a scraper on a particular job.

The optimum haul distance for small- and medium-size scrapers is 1 km or less. The bowl capacity of scrapers vary from 20 m³ to 40 m³.

The basic operating parts of a scraper are:

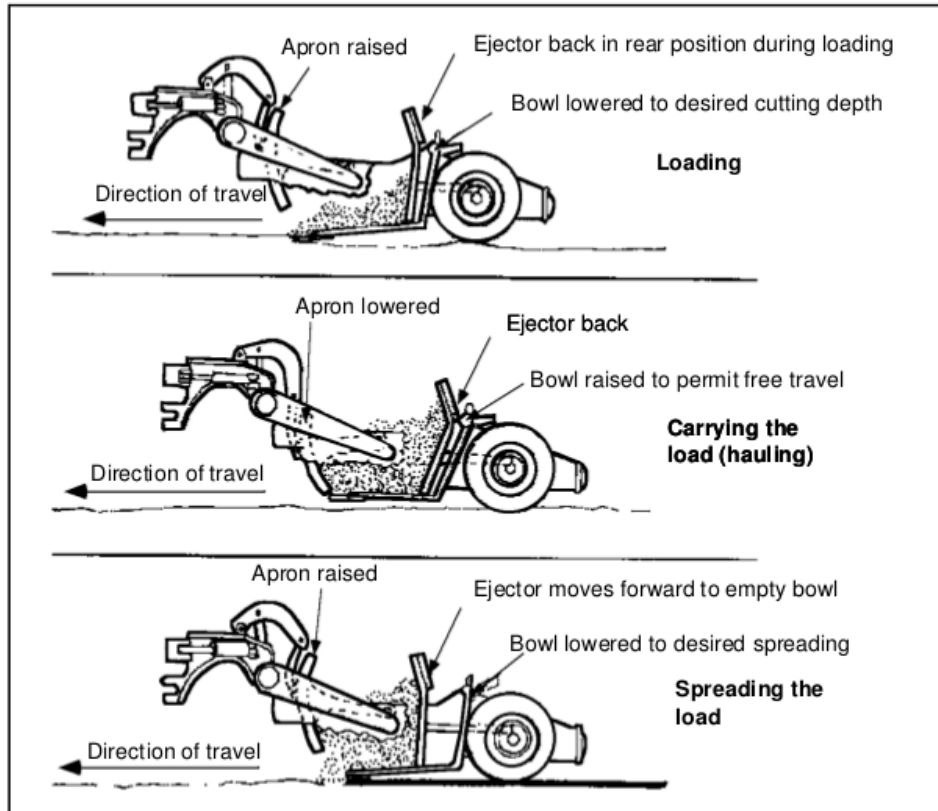


Bowl: The bowl is the loading and carrying component. It has a cutting edge, which extends across the front bottom edge. Lower the bowl until the cutting edge enters the ground for loading, raise it for carrying, and lower it to the desired lift thickness for dumping and spreading.

Apron: The apron is the front wall of the bowl. It is independent of the bowl and, when raised, it provides an opening for loading and spreading. Lower the apron during hauling to prevent spillage.

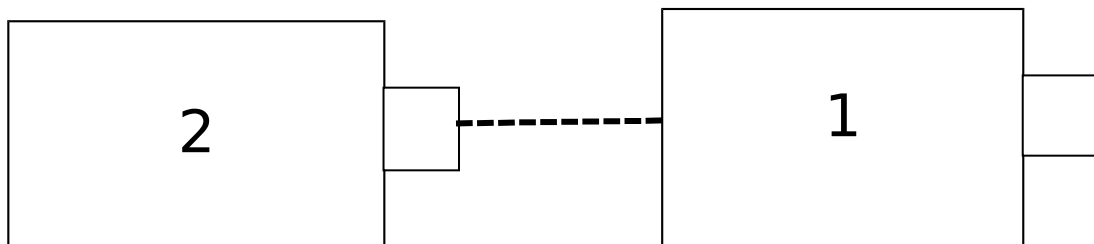
Ejector: The ejector is the rear wall of the bowl. Keep the ejector in the rear position when loading and hauling materials. Activate the ejector to move forward during spreading to provide positive discharge of materials.

The basic cutting sequence is shown in Figure below:



Push-Pull operation

In this operation, no dozer is required or used pushing the scraper. Instead two scrapers help each other during this phase.



These two machines will be connected by push pull hitch. The front machine is loading while another is pushing this as it has extra power because it is not cutting. Once the 1st machine is loaded and bowl has been taken up then the 2nd machine comes into action for cutting and loading while it is being assisted by the 1st machine through pulling. This way both the machines are fully loaded. Then they are separated by dis-engaging the push pull rope and move to the dumping ground independently.

During the time, the two machines are hooked together; they can't make any angular movement w.r.t to each other.

Advantage – Pusher dozer is not required.

Disadvantage – Strict scheduling of scrapers is required. Minimum area of cut length is

comparatively larger. The scraper is required to follow straight path during cutting and loading. Cutting and loading time is more as compared to pusher dozer operation.

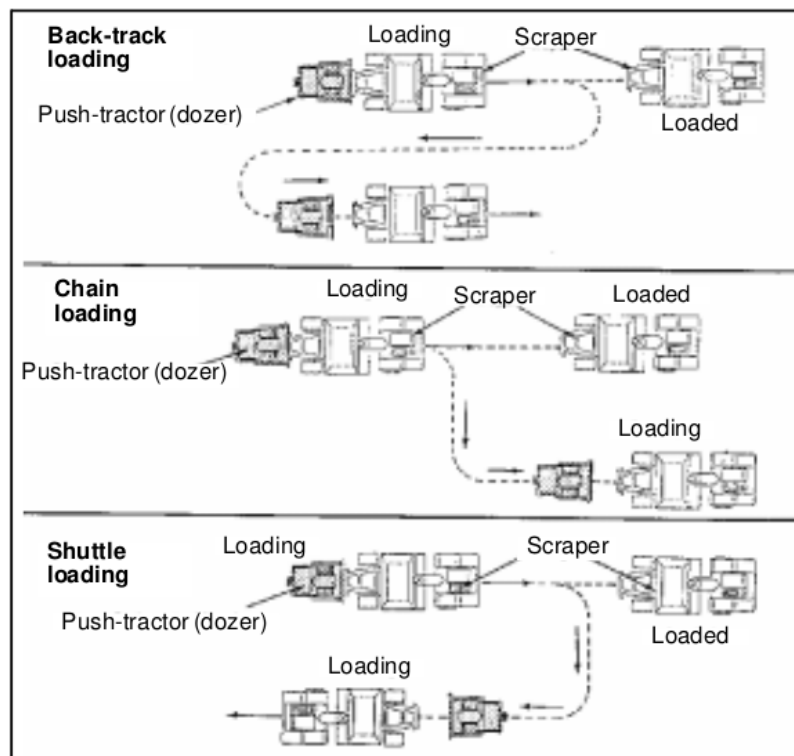
Pusher Dozer operation

When a scraper (standard or tandem) starts cutting and loading, a pusher dozer engages its push blade to push block scraper and supplies the required additional power during the operation of this cutting and loading phase.

The pushing can be done in following three systems

- a) Back-track loading is most commonly used since it permits all scrapers to load in the same general area. It is the slowest of the three methods because of the additional pusher travel distance.
- b) Chain loading is suitable for a long, narrow cut area.
- c) Shuttle loading requires two separate fill areas for efficient operations.

Modes of pushing scrapers by pusher dozers



Advantage

Strict scheduling of scraper is not required. As soon as the scraper comes to the working area, it is engaged with the dozer.

Minimum area of cut length is comparatively shorter. The scraper may not be required to follow straight path during cutting and loading. Cutting and loading time will be less as compared to push pull operation.

Disadvantage – Pusher dozer is required.